

Design and Implementation of a Gesture-Controlled Wheelchair Using Distributed Embedded Control

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Abstract—Assistive mobility devices such as powered wheelchairs are essential for individuals affected by lower-limb paralysis. Conventional joystick-based wheelchairs require precise finger coordination and are often costly. This paper presents a low-cost gesture-controlled wheelchair based on a distributed embedded control architecture. A wrist-mounted transmitter streams raw inertial sensor data to a receiver unit mounted on the wheelchair. All intelligence and safety logic are implemented on the receiver. The system supports proportional speed control, diagonal motion using dual-axis gestures, smooth acceleration via rate-limited velocity control, and immediate fail-safe stopping under communication failure. The proposed design emphasizes safety, modularity, and scalability.

Index Terms—Gesture control, Assistive technology, Wheelchair automation, Embedded systems, Safety-critical control

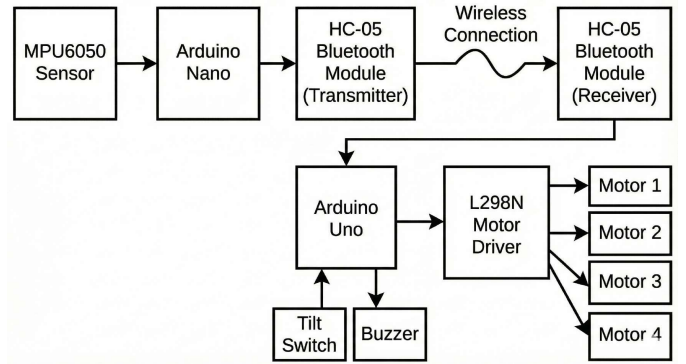


Fig. 1. High-level block diagram of the proposed system

I. INTRODUCTION

Mobility impairment significantly affects the independence and quality of life of individuals suffering from lower-limb paralysis. While powered wheelchairs provide an effective solution, joystick-based interfaces require fine motor skills that are not available to all users. Gesture-based control using inertial sensors offers a more intuitive and natural alternative.

Many existing gesture-controlled wheelchairs embed control and decision logic within the sensing unit, leading to limited safety guarantees and poor fault isolation. This work proposes a distributed embedded control system in which sensing and actuation are strictly separated. The transmitter performs only data acquisition and transmission, while all control, safety, and motion logic reside on the receiver.

II. SYSTEM ARCHITECTURE

A. Overall Architecture

The system consists of two independent modules:

- **Transmitter Unit (TX):** Wrist-mounted sensor module
- **Receiver Unit (RX):** Wheelchair-mounted control and actuation module

All intelligence and safety logic are implemented on the receiver, ensuring fail-safe operation.

III. TRANSMITTER UNIT DESIGN

The transmitter unit consists of an Arduino microcontroller interfaced with an MPU6050 inertial measurement unit. Its responsibilities are deliberately minimal:

- Acquisition of raw gyroscope data
- Sequence numbering of packets
- CRC generation for data integrity
- Periodic transmission at 50 Hz

No gesture recognition or control logic is implemented on the transmitter.

A. Packet Structure

Each transmitted packet is a fixed-length binary frame:

$$\text{Packet} = \{S_n, G_x, G_y, G_z, \text{CRC}\} \quad (1)$$

where S_n is the sequence number and G_x , G_y , and G_z are raw gyroscope measurements.

IV. RECEIVER UNIT DESIGN

The receiver unit validates incoming packets, computes motion commands, and drives the wheelchair motors. It enforces strict fail-safe behavior.

A. Communication Integrity

A packet is considered valid only if:

$$CRC_{rx} = CRC_{calc} \quad (2)$$

and the packet sequence is preserved:

$$S_n = S_{n-1} + 1 \quad (3)$$

Additionally, a communication timeout T_{timeout} ensures stopping on data loss.

B. Fail-Safe Philosophy

The proposed system follows a *fail-safe dominant* design philosophy, where the absence of valid control input is interpreted as a command to stop. This approach is commonly adopted in safety-critical embedded systems.

V. MOTION CONTROL MODEL

A. Gesture-to-Velocity Mapping

Let P represent pitch (forward/backward tilt) and R represent roll (left/right tilt). These components are derived as:

$$P = k_p \cdot G_y \quad (4)$$

$$R = k_r \cdot G_x \quad (5)$$

where k_p and k_r are scaling constants.

B. Differential Drive Mixing

The left and right wheel velocities are computed as:

$$V_L = P - R \quad (6)$$

$$V_R = P + R \quad (7)$$

This formulation enables diagonal motion when pitch and roll gestures are applied simultaneously.

VI. SMOOTH ACCELERATION CONTROL

Direct application of target velocities may cause jerky motion and mechanical stress. To mitigate this, rate-limited velocity control is employed:

$$V_{cur}(k+1) = V_{cur}(k) + \text{sat}(V_{tar} - V_{cur}) \quad (8)$$

where $\text{sat}(\cdot)$ limits the maximum velocity change per control cycle. During emergency conditions, acceleration smoothing is bypassed and motor outputs are immediately forced to zero.

VII. ENABLE CONTROL AND SAFETY HANDLING

A hardware enable button allows the user or caretaker to activate or deactivate motion. When disabled, all motor commands are ignored.

Fail-safe stopping is triggered under the following conditions:

- CRC mismatch
- Sequence number mismatch
- Communication timeout
- Manual disable

VIII. EXPERIMENTAL EVALUATION

The system was evaluated under continuous motion, diagonal movement, and induced communication failure scenarios. Smooth velocity transitions were observed during normal operation, while immediate stopping occurred under packet loss or corruption, validating the safety mechanisms.

IX. CONCLUSION

This paper presented a distributed, safety-oriented gesture-controlled wheelchair system. By separating sensing from control and enforcing strict receiver-side safety logic, the system achieves intuitive control, smooth motion, and robust fail-safe operation. The architecture provides a strong foundation for future intelligent assistive mobility systems.

NOMENCLATURE

Symbol	Description
G_x, G_y	Raw gyroscope measurements
P	Pitch component
R	Roll component
V_L, V_R	Left and right wheel velocities
k_p, k_r	Scaling constants
T_{timeout}	Communication timeout

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